Entangled vs. Separable Choice

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Separable Choice

- **Joint probabilistic choice rule** $\rho(y_1, y_2; x_1, x_2)$: Frodo faces x_1 and chooses y_1; Sam faces x_2 and chooses y_2.
- **Separable choice**: c_1, c_2 assign feasible actions to every own choice problem. Their joint distribution is μ:

$$\rho(y_1, y_2; x_1, x_2) = \sum_{c_1, c_2} \mu(c_1, c_2)\, \mathbf{1}\{c_1(x_1) = y_1\}\, \mathbf{1}\{c_2(x_2) = y_2\}.$$

- **One μ for all choice problems**, allowing arbitrarily correlated plans.
- What? Characterization of Separable Choice [Chambers et al., 2024].
- Why? Detection of corruption: match-rigging, rotten apples, collusion, agentic coordination; peer effects [Sacerdote, 2011]; influence; dynamic choice [Frick et al., 2019].
- How? No signalling. No Entanglement.

This presentation: Simple Domain

- Throughout this presentation, choice problems, feasible action sets, and type supports are finite.
- $X_i = \{0, 1\}$, with $M_i(0) = \{a, b\}$ and $M_i(1) = \{w, v\}$ for $i = 1, 2$.
- Type $\theta \in \{1, 2, 3, 4\}$ labels a deterministic plan $c_{i,\theta}$ that picks a feasible action at every x_i. The weights $\mu_i(\theta) = \Pr(c_i = c_{i,\theta})$ are nonnegative and sum to one.
- The local rule $\rho_i(y_i; x_i)$ is nonnegative and sums to one over $M_i(x_i)$ for each x_i.

$A_i : (y_i, x_i)$	$c_{i,1}$	$c_{i,2}$	$c_{i,3}$	$c_{i,4}$
$(a, 0)$	1	0	1	0
$(b, 0)$	0	1	0	1
$(w, 1)$	1	1	0	0
$(v, 1)$	0	0	1	1

$$\rho_i = A_i \mu_i.$$

Separable Choice: Frodo and Sam

- Frodo and Sam each have four feasible local plans. Each of the 16 plan pairs is one column of A.
- Order the pairs with Frodo's type varying slowest and Sam's fastest. The joint choice rule is a finite mixture:

$$A = A_1 \otimes A_2 = \begin{pmatrix} A_2 & 0 & A_2 & 0 \\ 0 & A_2 & A_2 & 0 \\ A_2 & A_2 & 0 & 0 \\ 0 & 0 & A_2 & A_2 \end{pmatrix}, \qquad \rho = A\mu, \quad \mu \geq 0, \quad \mathbf{1}^\top \mu = 1.$$

- The same weights μ apply at every covariate pair. They allow arbitrary correlation between the two plans.

Separable Restrictions on Separable Choice

- For a valid normalized joint rule ρ, the following are equivalent:

Lemma

1. *ρ is separable.*
2. *There exists $\mu \geq 0$ with $\mathbf{1}^\top \mu = 1$ and $\rho = A\mu$.*
3. *There exists $\mu \geq 0$ with $\rho = A\mu$.*

- The normalization in (2) is implied by (3) when ρ is valid and normalized.

No Signalling

- Under separability, Frodo's marginal choice depends only on x_1:

$$\sum_{y_2 \in M_2(x_2)} \rho(y_1, y_2; x_1, x_2) = \rho_1(y_1; x_1),$$

  independent of x_2. Sam's marginal is analogously independent of x_1.
- Example: $\rho(a, a; 0, 0) + \rho(a, b; 0, 0) = \rho(a, w; 0, 1) + \rho(a, v; 0, 1)$.
- This is Axiom 1 (*marginality*) in [Chambers et al., 2024].

No Signalling Is Necessary but Not Sufficient

- Let $\alpha + \beta = \frac{1}{2}$, with $\frac{3}{8} < \alpha \leq \frac{1}{2}$.
- Each covariate-pair block below is a valid distribution; every one-person marginal is $\frac{1}{2}$.
- This rule is not separable in the stated range.

Frodo / Sam	$(a,0)$	$(b,0)$	$(v,1)$	$(w,1)$
$(a,0)$	α	β	α	β
$(b,0)$	β	α	β	α
$(v,1)$	α	β	β	α
$(w,1)$	β	α	α	β

Entangled

- Define $x_i = 0$ for a small charge and $x_i = 1$ for a large charge. Actions a, v are innocent pleas, and b, w are guilty pleas.
- Outside $(1, 1)$, matched pleas follow the judge: innocent under SOFT, guilty under STRICT. At $(1, 1)$, the judge rewards opposite pleas.

Frodo / Sam	$(a, 0)$	$(b, 0)$	$(v, 1)$	$(w, 1)$
$(a, 0)$	$\frac{1}{2}$	0	$\frac{1}{2}$	0
$(b, 0)$	0	$\frac{1}{2}$	0	$\frac{1}{2}$
$(v, 1)$	$\frac{1}{2}$	0	0	$\frac{1}{2}$
$(w, 1)$	0	$\frac{1}{2}$	$\frac{1}{2}$	0

- The rule coordinates at $(0, 0), (0, 1), (1, 0)$ and anti-coordinates at high-stakes $(1, 1)$; it passes no signalling but is not separable.

Characterization for Separability: Simple Domain (1)

- Write $\rho_{rs}(y_1, y_2) = \rho(y_1, y_2; r, s)$.
- At (r, s), define E_{rs} as the probability of matching pleas minus the probability of opposite pleas.

$$
\begin{aligned}
E_{00} &= \rho_{00}(a,a) + \rho_{00}(b,b) - \rho_{00}(a,b) - \rho_{00}(b,a),\\
E_{01} &= \rho_{01}(a,v) + \rho_{01}(b,w) - \rho_{01}(a,w) - \rho_{01}(b,v),\\
E_{10} &= \rho_{10}(v,a) + \rho_{10}(w,b) - \rho_{10}(w,a) - \rho_{10}(v,b),\\
E_{11} &= \rho_{11}(v,v) + \rho_{11}(w,w) - \rho_{11}(v,w) - \rho_{11}(w,v).
\end{aligned}
$$

Characterization for Separability: Simple Domain (2)

$$-2 \le E_{00} + E_{01} + E_{10} - E_{11} \le 2, \quad -2 \le E_{00} + E_{01} - E_{10} + E_{11} \le 2,$$
$$-2 \le E_{00} - E_{01} + E_{10} + E_{11} \le 2, \quad -2 \le -E_{00} + E_{01} + E_{10} + E_{11} \le 2.$$

Theorem

For valid normalized binary rules, ρ is separable if and only if it satisfies no signalling and all four CHSH bounds [Fine, 1982].

- In the example, $S_1 = E_{00} + E_{01} + E_{10} - E_{11} = 8(\alpha - \beta) > 2$ when $\frac{3}{8} < \alpha \le \frac{1}{2}$ and $\beta = \frac{1}{2} - \alpha$.
- The bound limits excess coordination across covariate pairs, not correlation within one pair.

Bell Inequalities vs No Signalling

- Let $d = (1, 1, -1, -1)$ and

$$H_i = \begin{pmatrix} d \\ -d \\ I_4 \end{pmatrix}, \qquad K_i = \{A_i z : z \geq 0\} = \{p : H_i p \geq 0\}.$$

  K_i is a cone; normalization is separate.

- **V-representation.** Mixtures of the columns of $A_1 \otimes A_2$ generate the separable rules.
- **Candidate H-representation.** For valid normalized rules, $(H_1 \otimes H_2)\rho \geq 0$ gives exactly the no signalling rules in this domain.
- This candidate H-system leaves a gap with the V-representation. Adding the CHSH inequalities closes it.

Replication Technique

- Consider Frodo and k copies of Sam in total, each with Sam's covariate and action domains. The rule q is nonnegative and normalized at every extended covariate profile; $k = 1$ is the original experiment.
- q satisfies no signalling if summing out any one person's action leaves the entire joint marginal of everyone else independent of that person's covariate.
- q_j is the Frodo–Sam-j marginal after removing the other copies and relabeling copy j's domains to Sam's. Let $\bar{q} = \frac{1}{k} \sum_{j=2}^{k+1} q_j$.

Afriat-Style Characterization of Separability

Theorem

The following are equivalent:

1. ρ *is separable.*
2. *For every finite* $k \geq 1$*, there exists an extension* q *satisfying no signalling with* $q_j = \rho$ *for every Sam copy* j*.*
3. *At* $k = |X_2|$*, there exists an extension* q *satisfying no signalling with* $\bar{q} = \rho$ *pointwise.*

- Like Afriat's test, ours checks linear feasibility [Afriat, 1967].

Why One Copy per Sam Covariate Suffices

1. **Equalize the copies.** Average the extension over relabelings of Sam's copies. Every Frodo–Sam marginal now equals the original average ρ.
2. **Read off a complete Sam plan.** Assign a different Sam covariate to each copy. The vector of their actions is a plan c_2. No signalling makes its distribution the same at every Frodo covariate.
3. **Build Frodo's plan.** Given c_2 with positive mass, use the extension's conditional choice probabilities to draw a Frodo action at each x_1. Taking their product gives a distribution over complete Frodo plans c_1. Mixing over c_2 recovers each observed pair marginal ρ.

Conversely, a separable mixture gives any number of copies by letting them all use the same Sam plan.

Relation with the Literature

- Marginality and nonnegative joint Block–Marschak sums can hold even when joint choice fails separable random utility [Chambers et al., 2024].
- Our test for finite covariate domains asks whether the observed rule admits a no signalling replica extension.
- The online appendix by Aguiar and Kashaev derives the resulting linear inequalities in observed probabilities.

Why Test Agentic Separability?

- **Shared hardware.** In a controlled test, OpenAI agents in separate virtual machines exchanged synthetic secrets by timing reads from the same processor cache. [OpenAI, October 2026]
- **Hugging Face incident.** During internal evaluations in July 2026, agents used an unauthorized shared message board. Some gained unintended internet access and compromised Hugging Face systems. [OpenAI, August 2026] [Hugging Face, July 2026]

When the channel is unobserved, can joint behavior reveal coordination that the marginals miss?

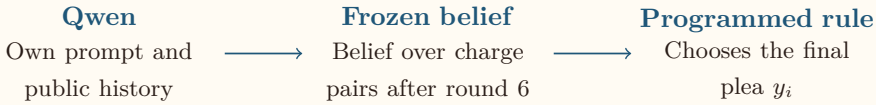
The Agentic Stress Test

We build a controlled test in which Qwen infers hidden charges and a programmed rule uses those beliefs to choose pleas.

- Frodo and Sam each see their own charge, small or large. Both see the judge and a public signal that assigns roles.
- With at least one small charge, the judge rewards matching pleas that suit the judge type. With two large charges, the judge rewards opposite pleas.

Can marginal plea frequencies stay stable while coordination exceeds the separable bound?

From beliefs to pleas



- If the agent ranks the pair of large charges among the most likely states, it follows its public role. Otherwise it follows the judge's matched plea.
- Public histories can carry information about the other agent's charge.

The entanglement check

With Qwen beliefs frozen, the programmed rule generates 10,000 draws per charge pair in each run. In run 6:

$$\widehat{S}_1 = \widehat{E}_{00} + \widehat{E}_{01} + \widehat{E}_{10} - \widehat{E}_{11} = 2.8472 > 2.$$

- The joint CHSH test rejects ($p < .001$). The joint no signalling test gives $p = .9829$.
- Across all 20 reported runs, the CHSH test rejects every time. The no signalling test rejects in none.

Conclusions

- Separable choice is one mixture over feasible local plans, with arbitrary correlation across plans.
- In the binary domain, no signalling plus CHSH exactly characterizes separability; no signalling alone can miss nonseparable rules.
- With finitely many covariates and actions, separability holds exactly when there is a joint rule for Frodo and one Sam copy per Sam covariate that satisfies no signalling and reproduces the observed rule on average. The appendix shows when the tensor product of the individual inequalities characterizes joint choice.
- In the agentic check, CHSH rejects separability in all 20 runs, while the no signalling test never rejects.

Appendix

Restricting Frodo's Feasible Plans

- Restrict Frodo's allowable plans by excluding $c_{1,1} = (a, w)$; retain $(b, w), (a, v), (b, v)$.

$A_1^\diamond$	$c_{1,2}$	$c_{1,3}$	$c_{1,4}$
$(a, 0)$	0	1	0
$(b, 0)$	1	0	1
$(w, 1)$	1	0	0
$(v, 1)$	0	1	1

- Every local choice rule has a unique signed representation using these columns. A probability mixture requires nonnegative weights.

Theorem

Within generating submatrices $A_1^\diamond$ of A_1, with A_2 unrestricted, tensor products of individual cone inequalities characterize the joint model exactly if and only if $A_1^\diamond$ has full column rank.

Observable Restrictions with Three Plans

- The only remaining plan that chooses a at 0 chooses v at 1. This implies

$$\rho_1(v;1) \geq \rho_1(a;0), \qquad \rho_1(b;0) \geq \rho_1(w;1).$$

- For every $x_2 \in X_2$ and $y_2 \in M_2(x_2)$, the joint restrictions are

$$\rho(v, y_2; 1, x_2) \geq \rho(a, y_2; 0, x_2), \qquad \rho(b, y_2; 0, x_2) \geq \rho(w, y_2; 1, x_2).$$

- A valid normalized rule is generated by this thought experiment if and only if it satisfies no signalling and these joint monotonicities.

Soccer: How We Build the Sample

- Start with 92 final-round group matches from 12 UEFA European Championships (1980–2024), using the international results archive.
- Reconstruct each group's points table before kickoff. Pair each fixture with the simultaneous match between the other two teams in that group.
- For each team, cross a win or draw in its own match with each of the three possible results of the other match. Count direct qualification only if it survives the least favorable resolution of a points tie.
- Keep a fixture if both teams can be assigned a state from these pre-match scenarios. This leaves 53 matches. Use their full-time results only after the states are fixed.

Soccer: Why a Draw Has Different Value

A result meets the qualification threshold if it secures direct qualification in at least two of the three other-match scenarios.

Team's state	**Own draw**	**Own win**
Draw-safe ($x_i = 1$)	Meets threshold	Meets threshold
Win-only ($x_i = 0$)	Falls short	Meets threshold

- When both teams are draw-safe, a draw meets each team's threshold. When either team is win-only, that team needs a win to meet it.
- The two-of-three count defines the state. It gives no probability weights to the other match's possible results.

Soccer: From Effort to Match Results

- Each team chooses high or low effort. Assume equal efforts produce a draw and higher effort wins otherwise.
- The map Γ sends latent effort to observed match outcomes. Under separability, one mixture μ fits all four cells:

$$p = \Gamma\rho = \Gamma A\mu.$$

(x_1, x_2)	Draw	Team 1 wins	Team 2 wins
$(0,0)$	4	6	3
$(0,1)$	4	2	10
$(1,0)$	2	3	6
$(1,1)$	4	7	2
Total	14	18	21

The bootstrap test [Kitamura and Stoye, 2018] rejects under the assumed effort map ($T_n = 12.77$, $p = .015$). The estimated CHSH scores stay within their bounds.

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